# CHILUKA RISHITH REDDY

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## CAREER OBJECTIVE

Electronics and Instrumentation Engineering student with a strong interest in industrial instrumentation, process monitoring, measurement systems, and automation. Seeking to apply knowledge of sensors, data acquisition, electronics, and signal processing to support efficient industrial operations, equipment monitoring, process reliability, and continuous improvement.

## EDUCATION

**B.Tech (Electronics & Instrumentation Engineering)** 2023 – 2027
*Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering &Technology*
(Pursuing 7th sem) CGPA:7.52

## CERTIFICATIONS

- Coursera Certification in **Google Data Analytics.**
- **NPTEL Certification** in The Joy of Computing with **Python with Elite**, 2025.
- **Presented a research paper** titled "Hybrid Statistical Physics- Informed Graph Neural Network (SP-GNN) for Robust Fault Detection and Uncertainty Quantification in Distributed Sensor Systems" at **ICAMSTA-2025, Osmania University.**

## SKILLS

- **Instrumentation & Measurement:** Sensors, Sensor Interfacing, Data Acquisition, Signal Conditioning
- **Electronics:** Analog Electronics, Digital Electronics, Circuit Fundamentals
- **Control & Signal Processing:** Control Systems, Digital Filtering, FFT, HR/HRV Analysis
- **Embedded Systems:** STM32, Arduino, ESP32, Microcontrollers, GPIO, ADC
- **Programming:** C, Embedded C, Python
- **Engineering Tools:** MATLAB, LabVIEW, STM32CubeIDE, Arduino IDE

## PROJECTS

**PPG-Based AF Detection on STM32 — Inter-Patient Validation**
Developing an STM32-based PPG signal acquisition and AF detection system incorporating preprocessing, HR/HRV analysis and lightweight classification, with patient-disjoint validation for evaluating generalization to unseen subjects.

**Hybrid Statistical-Physics Informed Graph Neural Network for Robust Fault Detection and Uncertainty Quantification in Distributed Sensor Systems**
Developed an unsupervised SP-GNN framework for sensor fault detection using the Intel Berkeley Research Lab dataset comprising **54 sensors and approximately 2.3 million readings**. Integrated graph-based sensor relationship modelling and physics-informed message passing to identify anomalous sensor behaviour, achieving a **mean reconstruction error of 0.028**, compared with 0.072 for GCN and 0.049 for Physics-GNN.

**Embedded EV Monitoring System using Arduino**
Developed an **Arduino-based embedded monitoring prototype** for real-time acquisition and monitoring of key EV parameters, including **battery level, vehicle speed, and temperature**. Implemented sensor interfacing and data acquisition with a software dashboard for real-time visualization and estimated driving-range monitoring.

## ACHIEVEMENTS

- **1st Prize – Patent Summit Competition:** Passive Alcohol Detection System for Vehicle Safety, National Workshop on Patent Analytics and Filing Framework, VNRVJIET
- Participated in **24-Hour INNO-VATHON 3.0** Hackathon organized by the Department of EIE, VNRVJIET, developing innovative **hardware-software integrated solutions**., VNR VJIET
- Participated in a **four-day hands-on Embedded Systems workshop at VNRVJIET**, gaining practical exposure to microcontrollers, sensor interfacing, embedded programming, and hardware–software integration.